

ICP Crisis/Herniation Quick Reference

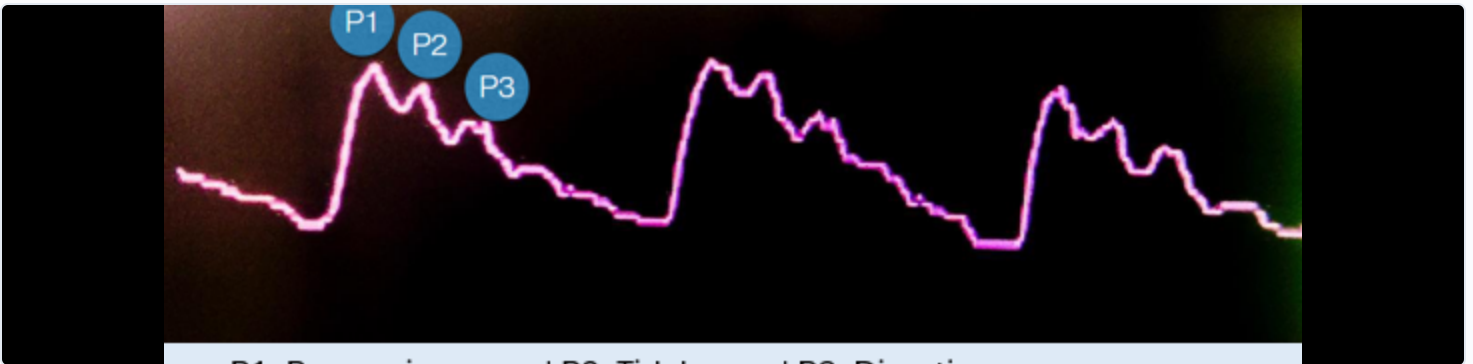
CLINICAL SIGNS OF HERNIATION

- **Motor decline:** Spontaneous GCS motor score decrease of ≥ 1 point.
- **Pupillary reactivity:** Decrease in pupillary reactivity (Neurological Pupil Index, NP_i < 3).
- **Asymmetry:** New pupillary asymmetry or unilateral dilation (ipsilateral mydriasis).
- **Focal deficit:** New focal motor deficit or abnormal posturing (decorticate / decerebrate).
- **Cushing's Triad (Late Sign):** Systolic hypertension, bradycardia, and irregular respirations.

MANAGEMENT

- **Hyperosmolar Therapies:**
 - **Mannitol 20% solution:** 1 g/kg IV bolus over 20–30 min. Must use in-line 0.22-micron filter to catch crystals.
 - **Hypertonic Saline (23.4% NaCl):** 30 mL IV bolus over 5–10 min. **Requires central venous line access.**
 - **Hypertonic Saline (3% NaCl):** 150–250 mL IV bolus over 15–20 min. Can be run via large peripheral IV in emergencies.
- **Hyperventilation:** Use strictly as short-term bridge therapy (target PaCO₂ 30–35 mmHg) to buy time for definitive surgical decompression or osmotherapy. Avoid prolonged use due to cerebral vasoconstriction and ischemia risks.

ICP WAVEFORM ANALYSIS



- **P1 (Percussion wave):** Arterial pulsation.
- **P2 (Tidal wave):** State of intracranial compliance (elastic reserve).
- **P3 (Dicrotic wave):** Venous pulsations.
- **Compliance States:**
 - **Normal Compliance:** $P1 > P2 > P3$ (elastic brain tissue easily cushions pulsations).
 - **Impaired Compliance / High ICP:** $P2 > P1$ (brain tissue reserve exhausted; high risk of herniation).